

Word Embeddings for Khmer Text

Mini Project 3 - Natural Language Processing

Skip-gram Model, Neural Language Model & Hyperparameter Tuning

1. Corpus & Preprocessing

The corpus "temples.txt" contains Khmer text from three Wikipedia articles about Angkor Wat, Banteay Srei, and Siem Reap province. Tokenization was performed with khmer-nltk, a library specialized for Khmer word segmentation. Words with frequency below 10 were filtered out.

Metric	Value	
Raw characters	45,920	
Raw tokens	11,521	
After freq >= 10 filter	6,241	
Vocabulary size	182	
Training pairs (window=4)	46,804	

The filtering preserves meaningful content words while removing rare terms and noise. With 182 unique words in the vocabulary, the resulting 46,804 training pairs provide sufficient data for learning distributional word representations.

2. Part I: Skip-gram Model

A skip-gram model with negative sampling was implemented in PyTorch. Separate center and context embedding matrices are trained to distinguish observed (center, context) pairs from random negative samples drawn from a unigram distribution raised to the 3/4 power.

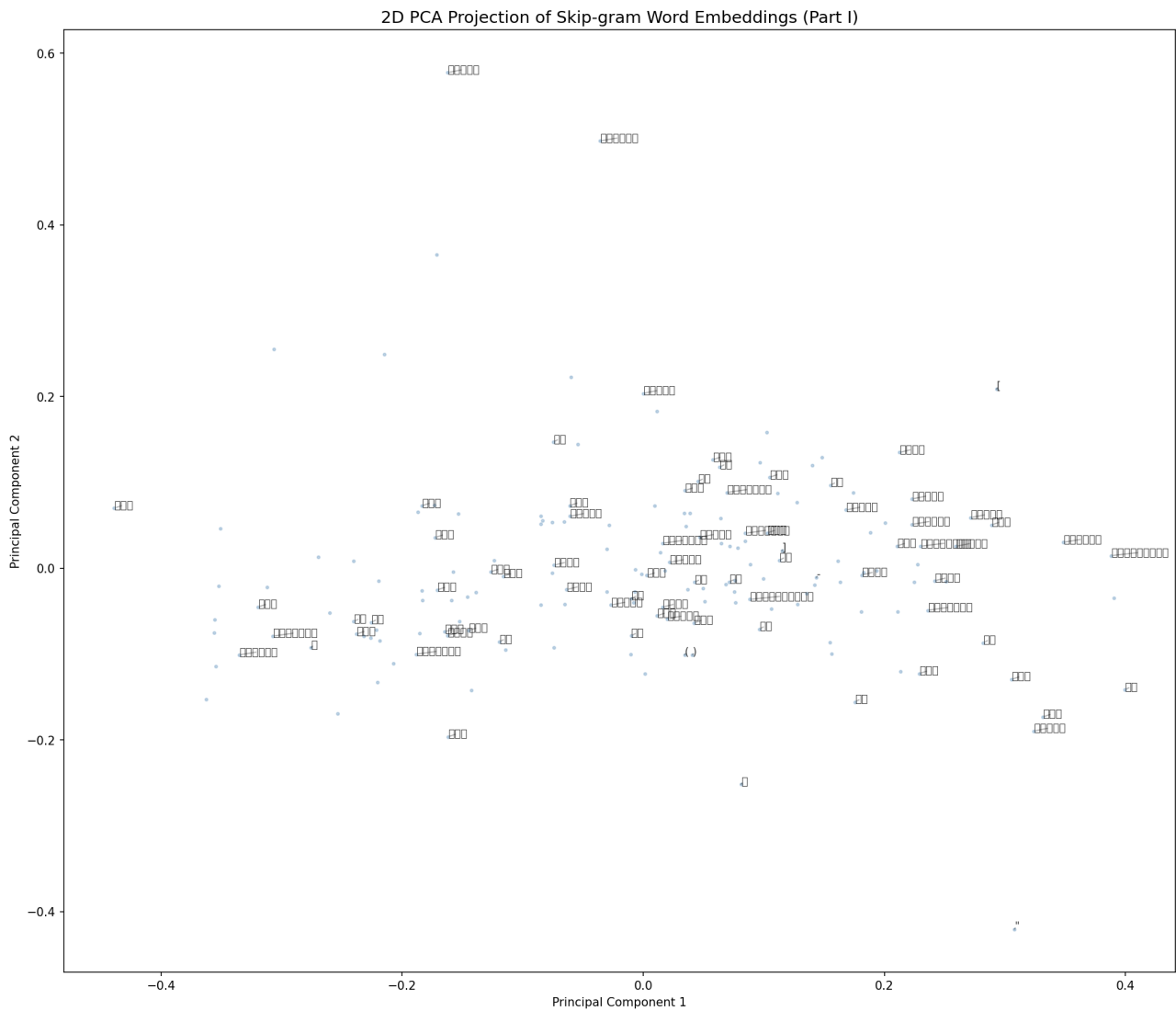
Parameter	Value	
Embedding dimension	50	
Context window	+/- 4	
Negative samples	2	
Training epochs	5	
Final loss	1.8832	
Parameters	18,200	

Word Similarity Results

The learned embeddings capture semantic relationships regarding temple domain.

3. Part II: PCA Visualization

Principal Component Analysis reduced the 50-dimensional embeddings to 2 components. The first two components explain 52.85% of the total variance.



4. Part III: Neural Language Model (Fixed Embeddings)

A neural language model predicts the next word from 5 previous words. The architecture concatenates the 50-dimensional pre-trained skip-gram embeddings (frozen) to form a 250-dimensional input, followed by a single hidden layer of 512 units.

Metric	Value		
Architecture	5x50 -> 512 -> 182		
Hidden activation	Sigmoid		
Training epochs	10		
Final loss	4.7324		
Perplexity	113.55		

5. Part IV: Learning Embeddings from Scratch

The retrained neural language model allows embeddings to be specialized for the next-word prediction task.

Metric	Fixed (Part III)	Scratch (Part IV)
Final loss	4.7324	3.7992
Perplexity	113.55	44.62
Embeddings init	Skip-gram (frozen)	Random (trainable)
PCA 2D variance	52.85%	8.57%

6. Hyperparameter Tuning

Dimension	Window	Neg	Pairs (K)	Loss	PCA 2D Var
30	2	1	23.4	1.3672	0.3909
30	2	2	23.4	1.8956	0.5486
50	6	2	70.2	1.8765	0.7104
100	6	1	70.2	1.3296	0.5051

7. Conclusion

This project successfully implemented a complete word embeddings pipeline for Khmer text.